

CITS4401

Software Requirements & Design

Group Project Phase 1

Group Number: Group 39

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Stakeholder Descriptions and User Stories

1. Stakeholder: Ethics Committee Members

1.1 Stakeholder Description

Ethics Committee Members are academic or professional staff appointed to any research ethics applications involving human participants. Their primary responsibility is to assess whether proposed research activities comply with institutional policies and ethical standards before approval can be granted. Their role involves evaluating application details such as research methodology, potential risks to participants, and supporting documentation (e.g., consent forms and risk assessments). They request clarifications or revisions if needed, participate in review discussions, and contribute to the final decision (e.g., approval, conditional approval, or rejection). Additionally, they are required to identify and declare any conflicts of interest and recuse themselves from reviews where necessary, ensuring fairness and integrity in the ethics review process.

Responsibilities:

- Review assigned research ethics applications, including all supporting documentation (e.g., consent forms, risk assessments), to ensure compliance with ethical standards and institutional policies
- Identify incomplete, unclear, or non-compliant applications and request clarifications or revisions from researchers
- Assess potential risks to human participants, including vulnerable groups, and ensure appropriate safeguards are in place
- Declare and manage conflicts of interest, including recusing themselves from applications where impartiality may be compromised
- Participate in collaborative review and discussion with other committee members for complex or high-risk applications
- Provide structured feedback and recommendations (e.g., approval, conditional approval, rejection) to support final decisions
- Communicate review outcomes and recommendations to administrative staff for further processing

Value/Benefits Expected from the System:

The system should reduce the current burden of locating and managing documents across fragmented email threads by providing a centralised platform for accessing all applications and supporting materials. It should support document version control with clear version histories and highlighted changes, enabling members to efficiently identify updates without re-reviewing entire

documents. Additionally, the system should provide timely notifications for application submissions, updates, and pending actions to improve responsiveness.

The system should also formalize conflict-of-interest declarations within the platform, replacing informal self-reporting processes with a structured and auditable mechanism. Overall, it should streamline access to all relevant application materials, improve transparency in the review process, and enhance the efficiency and consistency of ethics evaluations.

1.2 User Stories — Ethics Committee Members

User Story 1: Reviewing an Assigned Application

As an Ethics Committee Member, I want to view all documents and supporting materials for applications assigned to me in one centralized location, so that I do not have to search through emails or shared folders to find the relevant files.

User Story 2: Identifying Changes in Revised Documents

As an Ethics Committee Member, I want the system to highlight what has changed between document versions when a researcher submits a revision, so that I can focus my review on the updated content without having to read the entire document again.

User Story 3: Requesting Clarification from a Researcher

As an Ethics Committee Member, I want to submit a structured clarification request through the system, so that my feedback is documented, consolidated, and forwarded to the researcher by the administrative staff in an organised manner.

User Story 4: Declaring a Conflict of Interest

As an Ethics Committee Member, I want to formally declare a conflict of interest for a specific application through the system and recuse myself from its review, so that the declaration is recorded in the audit trail and the application can be reassigned appropriately.

User Story 5: Viewing Research Team Membership Before Review

As an Ethics Committee Member, I want to see the full list of researchers and research topics associated with an application before I begin reviewing it, so that I can identify any conflicts of interest prior to engaging with the application content.

User Story 6: Receiving Notifications of Application Updates

As an Ethics Committee Member, I want to receive a notification in the system when an application I am reviewing is updated or a revised document is submitted, so that I am aware of changes without having to monitor my email inbox.

User Story 7: Collaborating with Other Reviewers

As an Ethics Committee Member, I want to view comments and discussion notes from other reviewers on the same application, so that I can consider their perspectives and reach a coordinated review outcome.

2 Stakeholder: Ethics Committee Chair

2.1 Stakeholder Description

The Ethics Committee Chair is the primary authority within the ethics review process, responsible for overseeing and coordinating the entire committee's activities. The Chair holds ultimate responsibility for making final decisions on all ethics applications, including approving, conditionally approving, or rejecting submissions. They ensure that the review process is conducted consistently and efficiently and coordinate the assignment of applications to committee members.

Responsibilities:

- Make final decisions on ethics applications, including approval, conditional approval, or rejection
- Coordinate the assignment of applications to appropriate committee members, particularly for complex or high-risk cases
- Oversee the committee's overall workload and ensure timely and efficient processing of applications
- Ensure that conflicts of interest are properly declared, managed, and resolved, including reassigning applications where necessary
- Maintain consistency, fairness, and adherence to institutional ethical standards across all reviews

Value/Benefits Expected from the System:

The system should provide the Chair with a clear, real-time overview of all active applications, including their status, assigned reviewers, and any declared conflicts of interest. It should support structured and formal recording of decisions, ensuring that all outcomes are captured within an auditable trail. Additionally, the system should reduce reliance on informal coordination between committee members by centralising workflows, communication, and reviewer management, thereby enabling the Chair to exercise more effective oversight and governance of the entire review process.

2.2 User Stories — Ethics Committee Chair

User Story 1: Recording a Final Decision on an Application

As an Ethics Committee Chair, I want to formally record an approval, conditional approval, or rejection decision within the system, so that the decision is documented in the audit trail and automatically communicated to the relevant parties.

User Story 2: Assigning Committee Members to Complex Applications

As an Ethics Committee Chair, I want to assign one or more committee members to review a specific application, so that complex or high-risk applications receive adequate reviewer attention, and this is recorded in the system.

User Story 3: Monitoring Overall Application Progress

As an Ethics Committee Chair, I want to view a dashboard showing all current applications, their statuses, assigned reviewers, and any overdue items, so that I can identify gaps and ensure the review process proceeds in a timely manner.

User Story 4: Managing Conflicts of Interest Across the Committee

As an Ethics Committee Chair, I want to be notified when a committee member declares a conflict of interest, so that I can reassign the application to another available member promptly.

User Story 5: Overseeing Amendment and Extension Requests

As an Ethics Committee Chair, I want to distinguish between amendment requests (which require fuller review) and extension requests (which require lighter approval), so that the appropriate level of scrutiny is applied and processing time is not wasted on unnecessary full reviews.

User Story 6: Receiving Alerts for Stalled or Overdue Applications

As an Ethics Committee Chair, I want to receive automated alerts when an application has not progressed within an expected timeframe — such as a reviewer not responding — so that I can intervene before delays become critical.

3 Stakeholder: Study Participants

3.1 Stakeholder Description

Study Participants are individuals whose data, welfare, and rights are central to the ethics review process. Although they are not typically direct users of the REMS, they are the primary subjects of ethical protection. The system exists, in part, to safeguard their interests by ensuring that all research involving human

participants complies with ethical standards, data protection requirements, and informed consent protocols. Their protection is embedded within the system through the review and validation of application content, particularly consent forms, participant information sheets, and risk assessments.

Responsibilities:

- Providing informed consent to participate in research (outside the system, but documented within it)
- Being protected by the system's enforcement of ethical standards before research commences
- Benefiting from the assurance that any research involving them has been properly reviewed and approved

Value/Benefits Expected from the System:

Study Participants benefit indirectly but critically from the REMS. The system ensures that no research involving human participants can proceed without appropriate ethics approval, thereby safeguarding participant welfare. It enforces strict handling of sensitive data through confidentiality and access control mechanisms, reducing the risk of misuse or exposure. Additionally, the system ensures that key participant-facing documents, such as consent forms and information sheets, are accurate, complete, and version-controlled at the point of approval. It also guarantees that any amendments to research activities are subject to further ethical review before implementation.

3.2 User Stories — Study Participants

Note: Study participants are indirect beneficiaries of the REMS rather than active system users. Their user stories reflect the system's obligations to protect their interests.

User Story 1: Ensuring Approval Before Research Commences

As a Study Participant, I want assurance that the research I am invited to join has received full ethics approval before any data collection begins, so that my welfare and rights are protected throughout the study.

User Story 2: Protection of Sensitive Personal Data

As a Study Participant, I want my personal data and any information I provide as part of a study to be kept confidential and accessible only to authorised personnel, so that my privacy is not compromised.

User Story 3: Reviewed and Up-to-Date Consent Documentation

As a Study Participant, I want the consent forms and participant information sheets presented to me to reflect the most current, ethics-approved version of the research protocol, so that I can make a fully informed decision about participating.

User Story 4: Protection Through Conflict-of-Interest Safeguards

As a Study Participant, I want the ethics review of the research I am involved in to be conducted by reviewers who have no conflicts of interest with the research team, so that the review outcome is impartial and my welfare is not compromised by biased decision-making.

User Story 5: Ethics Re-Review Upon Significant Amendment

As a Study Participant, I want any significant changes to a research study I am participating in to be reviewed and re-approved by the ethics committee before being implemented, so that changes in methodology or risk level do not expose me to unanticipated harm.

4 Stakeholder: Researchers

4.1 Stakeholder Description

Researchers (including Honours, Masters, and PhD students, as well as academic staff) are the primary initiators of research activities. They are responsible for preparing and submitting ethics applications, including research methodologies, participant information, and supporting documentation. Throughout the process, they may need to respond to feedback, revise their applications, and monitor progress until a final decision is reached.

Responsibilities:

- Prepare and submit ethics applications and supporting documents (e.g., consent forms, risk assessments)
- Provide precise research descriptions, methodologies, and participant information for committee evaluation
- Respond to clarification requests and formally submit revised applications when required by the ethics committee
- Monitor the real-time status and lifecycle progression of submitted applications within the system
- Ensure all submitted information complies with institutional and ethical standards

Value/Benefits Expected from the System:

- **Real-time Status Tracking:** Researchers can monitor their application progress through a centralised dashboard. This removes the need for manual email follow-ups, as the system provides live updates on workflow stages as they happen
- **Centralised Source of Truth:** The system acts as a single integrated repository for all submissions. This centralises files and communications, which helps avoid the version confusion often caused by email chains
- **Integrity of Document Versioning:** The system uses a versioning engine to auto-archive old drafts and mark the 'Current Version'. This ensures reviewers always access the correct files and makes it easier to track document changes

4.2 User Stories — Researchers

User Story 1: As a researcher, I want to upload and update supporting documents in the central repository. The system should automatically timestamp these files and highlight the current version, so that the committee always reviews the correct documentation.

User Story 2: As a researcher, I want to view the real-time status of my application on a personal dashboard, where the system displays the current workflow stage (e.g., submitted, under review, revision required), so that I have complete visibility into my application's progress without manual follow-up.

User Story 3: As a researcher, I want to receive structured feedback through the system and submit revised versions within the same interface, and saves all changes in a revision history. This way, all feedback and updates are formally documented and easy to find.

User Story 4: As a researcher, I want to record the roles of all team members within the application metadata, where the system captures these details for formal review, so that institutional compliance regarding research team responsibilities is fully satisfied.

5 Stakeholder: Administrative Staff

5.1 Stakeholder Description

Administrative staff act as coordinators of the ethics review process. They are responsible for performing initial completeness checks, tracking application progress, managing communication between stakeholders, and ensuring the workflow operates efficiently. They play a central role in maintaining oversight of

all active applications and ensuring that processes are consistent and compliant with institutional requirements.

Responsibilities:

- Perform rigorous completeness checks on newly submitted applications and all associated supporting documentation
- Proactively track, update, and manage application statuses across the entire review lifecycle
- Facilitate professional communication flows between researchers and ethics committee members
- Coordinate the overall workflow, including routing applications for review and forwarding final outcomes
- Maintain detailed, searchable records to ensure absolute compliance with institutional ethical policies

Value/Benefits Expected from the System:

- **Unified Workflow Oversight:** The system provides an aggregated master dashboard to show all active applications, reviewer assignments, and deadlines. This allows staff to monitor the review process effectively and keep tasks on track
- **Workflow Automation and Standardization:** By utilizing automated validation logic and digital completeness checklists, the system enforces process consistency, ensuring no application proceeds to committee review until it meets established requirements
- **Immutable Auditability:** By logging every interaction including timestamped status changes, document uploads, and revision requests within a secure, system-wide database, the system ensures total transparency and simplifies compliance reporting

5.2 User Stories — Administrative Staff

User Story 1: As an administrative staff member, I want to use a digital checklist to verify applications, where the system validates the presence of mandatory documents before allowing a submission to proceed, so that I can ensure only high-quality, complete applications enter the review queue.

User Story 2: As an administrative staff member, I want to view all active applications via a master dashboard queue, where the system categorizes applications by status, submission date, and assigned reviewer, so that I can

monitor overall efficiency and prioritize urgent tasks.

User Story 3: As an administrative staff member, I want to issue formal revision requests via the system's communication module, where the system automatically updates the application status to 'Awaiting Revision', so that all feedback is formally logged and traceable.

User Story 4: As an administrative staff member, I want to update application statuses through the system's workflow control panel. The system triggers real-time updates and notifications to researchers. This keeps everyone informed and avoids confusion about the current status of an application.

User Story 5: As an administrative staff member, I want to filter stalled or overdue applications via the system reporting tool, where the system highlights items that have exceeded expected time limits, so that I can escalate cases to ensure timely processing.

User Story 6: As an administrative staff member, I want to configure user access levels within the system. The system enforces permission boundaries based on user roles, which ensures that sensitive research data is only accessed by authorized personnel, maintaining both confidentiality and data integrity.

Interview Report

Venue: EZONECENT 204A Meeting Pod

Date: 31/3/2026

Facilitator: Altong Won

Attendance:

- Kunyu He
- Dyutiman Shiv
- Sandra Franco Pynadath
- Yuxuan Xi

Stakeholder role: Administrative Staff

Question 1

When you check an application, what do you look for to decide whether it is complete? Does this differ depending on the type of research? What exactly happens if an application is deemed incomplete? Do you return it entirely to the researcher, or is it put 'on hold' while you request the missing documents?

Response: There is a checklist, but it is not updated or universal. It is passed out by anyone in the same department. They check if they have consent forms and assessments. For example if they deal with sensitive data, there will be additional forms for that. In case anything is missing, they will email the researcher themselves.

Key Insight: Completeness checks are currently inconsistent and heavily decentralized. Additional documentation depends entirely on the nature of the data (e.g., sensitive data requires more forms). Missing items are handled via direct ad-hoc emails rather than a formalized system status.

How did my understanding change: I initially assumed there was a strict, standardized checklist that administrative staff used for all applications. I now understand that checks are department-specific, informal, and that applications are not formally 'returned' but rather manually chased up via email.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must provide a dynamic checklist for researchers

during submission, automatically prompting for additional documents (like sensitive data forms) based on answers provided in the base application.

- Requirement: The system must allow administrative staff to flag specific missing documents, automatically pausing the application status to 'Incomplete - Pending Documents' and generating an automated email/notification to the researcher.

Question 2

How do you currently track the status of applications, and what problems do you encounter in this process?

Response: A spreadsheet is kept and is shared with everyone on the admin team. Each row is an application. Columns present submission date, committee member assigned, status. A pain point is that when not knowing who has updated it, they sometimes overwrite each other. The committee members decide the assignment on their own.

Key Insight: The current tracking system (a shared spreadsheet) suffers from severe concurrency issues (users overwriting each other's data) and lacks an audit trail. Furthermore, committee members self-assign applications rather than having them formally delegated by administrators.

How did my understanding change: I previously assumed that administrative staff manually assigned applications to reviewers based on workload. I now see that it is a 'pull' system where committee members select their own assignments, leading to coordination clashes.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must support concurrency control to prevent data from being overwritten by multiple administrative or committee users simultaneously.
- Requirement: The system must feature a centralized dashboard of unassigned applications, allowing committee members to securely 'claim' or self-assign applications.

Question 3

When an application has issues or needs revision, how do you communicate this to the researcher? Who is responsible for initiating that communication?

Response: The committee member assigned will decide if it needs revision. Then the admin staff will take it up to the researcher and ask them to submit the revised documents. All this communication is done via email.

Key Insight: There is a distinct multi-step handoff for revisions: the assigned committee member identifies the issue, but the administrative staff acts as the intermediary who actually emails the researcher.

How did my understanding change: I originally thought reviewers might communicate directly with researchers to save time. It is now clear that administrators act as a mandatory communication buffer between the Ethics Committee and the researchers.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must allow a committee member to flag an application as 'Requires Revision' and attach feedback, which is then routed into a queue for administrative staff to review and formally issue to the researcher.
- Assumption: Committee members prefer to remain somewhat insulated from direct back-and-forth negotiations with researchers.

Question 4

How are applications currently assigned to committee members? Do administrators manually select reviewers based on their expertise and workload, or are there strict university rules (e.g., based on risk level) that dictate exactly who must review it?

Response: This is internally decided between the committee members. It depends on factors such as who is available and is decided in a group and the chair gives the final approval. If the submission is important the chair looks over it.

Key Insight: Review assignment is a collaborative, consensus-based process rather than a rigid automated one. It requires final approval from the Committee Chair, who also personally oversees high-priority or high-risk applications.

How did my understanding change: I had assumed a simple round-robin or manual admin assignment. Instead, I now recognize the workflow requires group visibility, consensus, and a specific escalation path to the Chair for final approval or complex cases.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must allow the committee to propose reviewer assignments, with a feature for the Committee Chair to officially approve or modify the final assignment.
- Requirement: The system must automatically flag high-risk or high-priority applications and notify the Committee Chair for direct oversight.
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Question 5

What kinds of reminders or notifications would be most helpful for you in managing your workload?

Response: Receiving a notification would be helpful, then a tool that can highlight what needs to be revised. So that the communication with committee members and admin staff is smoother and a notification if the researcher who has been notified that a revision needs to happen but has not done it.

Key Insight: Users need targeted, automated notifications. Specifically, they need tools to highlight the exact location of required revisions within documents, and automated alerts to track overdue responses from researchers.

How did my understanding change: While I knew basic status updates were needed, I did not realize the critical need for granular document highlighting and 'timeout' notifications to prevent applications from stalling indefinitely.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must include an automated reminder feature that notifies researchers if a requested revision remains unsubmitted after a configured number of days.
- Requirement: The system must allow reviewers to digitally highlight, annotate, or directly link feedback to specific sections or documents within the application.

Question 6

In your view, what information should different roles (e.g., researchers, administrators, committee members) be permitted to access?

Response: If you are not related to the application then you can view basic information but the person involved should be able to see for transparency.

Key Insight: Strict confidentiality does not mean complete invisibility. Unassigned

committee members still need access to basic metadata, while full application details should be restricted only to actively involved individuals.

How did my understanding change: I initially believed that if a committee member was not assigned to an application, they should not see it at all. I now understand that a basic read-only summary view is required for broader committee transparency.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must implement Role-Based Access Control (RBAC) where unassigned staff/committee members can view high-level metadata (e.g., application title, status, assigned reviewer) but cannot access sensitive attached documents or detailed notes.

Question 7

How do you currently manage different versions of documents, and what problems does this cause?

Response: Use the latest email as the most current version, and I will keep it locally as well. The latest email will be forwarded to the committee members. Currently it is a mess and creates a lot of confusion.

Key Insight: Reliance on email chains and local hard drives means there is no single source of truth. Users treat the 'latest email' as the current version, causing massive confusion and fragmentation during the review process.

How did my understanding change: I knew versioning was a problem, but I did not realize reviewers were downloading local copies and forwarding them manually. The risk of someone reviewing an outdated document is incredibly high.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must serve as the single centralized repository for all application files, entirely eliminating the ability (or need) to submit documents via email.
- Requirement: The system must automatically manage document versioning, clearly displaying the most 'current' version while archiving older submissions under a unified history tab.

Question 8

When an approved application needs modification or an extension, how is this currently handled? What would make that process easier?

Response: If an already approved application needs modification or extension, then we would treat it as a new application.

Key Insight: There is currently no streamlined 'amendment' workflow. Extensions or modifications are forced through the exact same pipeline as brand-new applications.

How did my understanding change: The initial project brief mentioned amendments, so I assumed there was a specific, lightweight process for them. Finding out they are treated as entirely new applications highlights a major area where the new system can improve efficiency.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must include a distinct 'Amendment/Extension' workflow that allows a researcher to clone or link an existing approved application, highlighting only the fields/documents that have changed, to expedite the review process.

Question 9

Do the forms applied for research differ between staff members, Honours students, Masters or a PhD student?

Response: The base application does not change but beyond that the forms differ due to the severity of the research based on ethics.

Key Insight: The core application process is role-agnostic. The complexity of the application is driven entirely by the risk profile and severity of the proposed research, not the seniority of the applicant.

How did my understanding change: I originally assumed that academic staff or PhD students might face more rigorous or different baseline forms compared to Honours students. I now know the system logic should scale based on research risk, not user title.

Resulting Refinements to Requirements or Assumptions:

- Assumption: The base application template remains identical for all researcher roles.

- Requirement: The system must use conditional logic to append required forms (e.g., high-risk assessment forms) based on the specific ethical risks selected in the base application.

Question 10

In a research where there are multiple researchers involved, are the forms individually filled or is it done as a group?

Response: In this situation it is filled by the head researcher and he will be dealing with the whole application process.

Key Insight: Despite research being collaborative, the administrative burden falls solely on a single 'head researcher' who acts as the primary point of contact for the entire process.

How did my understanding change: I thought co-researchers might need joint editing privileges or required individual sign-offs in the system. It turns out a single principal investigator handles everything.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must allow one designated Principal Investigator (Head Researcher) to own the submission, receive all notifications, and execute revisions.
- Assumption: While co-researchers must be listed in the application for ethical compliance, they do not strictly require system accounts or edit permissions for the application to proceed.

Question 11

Is there any scenario where you might need to go through any previous documents that have been set?

Response: Just to be on the safer side, having previous documents to view helps during collaborative tasks. To verify changes.

Key Insight: Previous versions of documents are not just kept for archiving/auditing purposes; they are actively used by committee members to verify that requested changes were actually made by the researcher.

How did my understanding change: I thought old versions were just stored passively for compliance. I now understand that reviewers need them easily

accessible during the active review process to perform side-by-side comparisons.

Resulting Refinements to Requirements or Assumptions:

- Requirement: The system must allow reviewers to seamlessly toggle between or view previous versions of an application alongside the current submission.
- Requirement: The system should ideally incorporate a 'diff' or change-tracking feature, or force researchers to provide a 'summary of changes' document to aid reviewers in verifying revisions.

Requirements Specification

Subset of System Functionality Chosen: Application Review, Assignment, and Revision Subsystem.

This subsystem handles the workflow from the moment an application is submitted, through committee assignment, the review and feedback process, and document versioning.

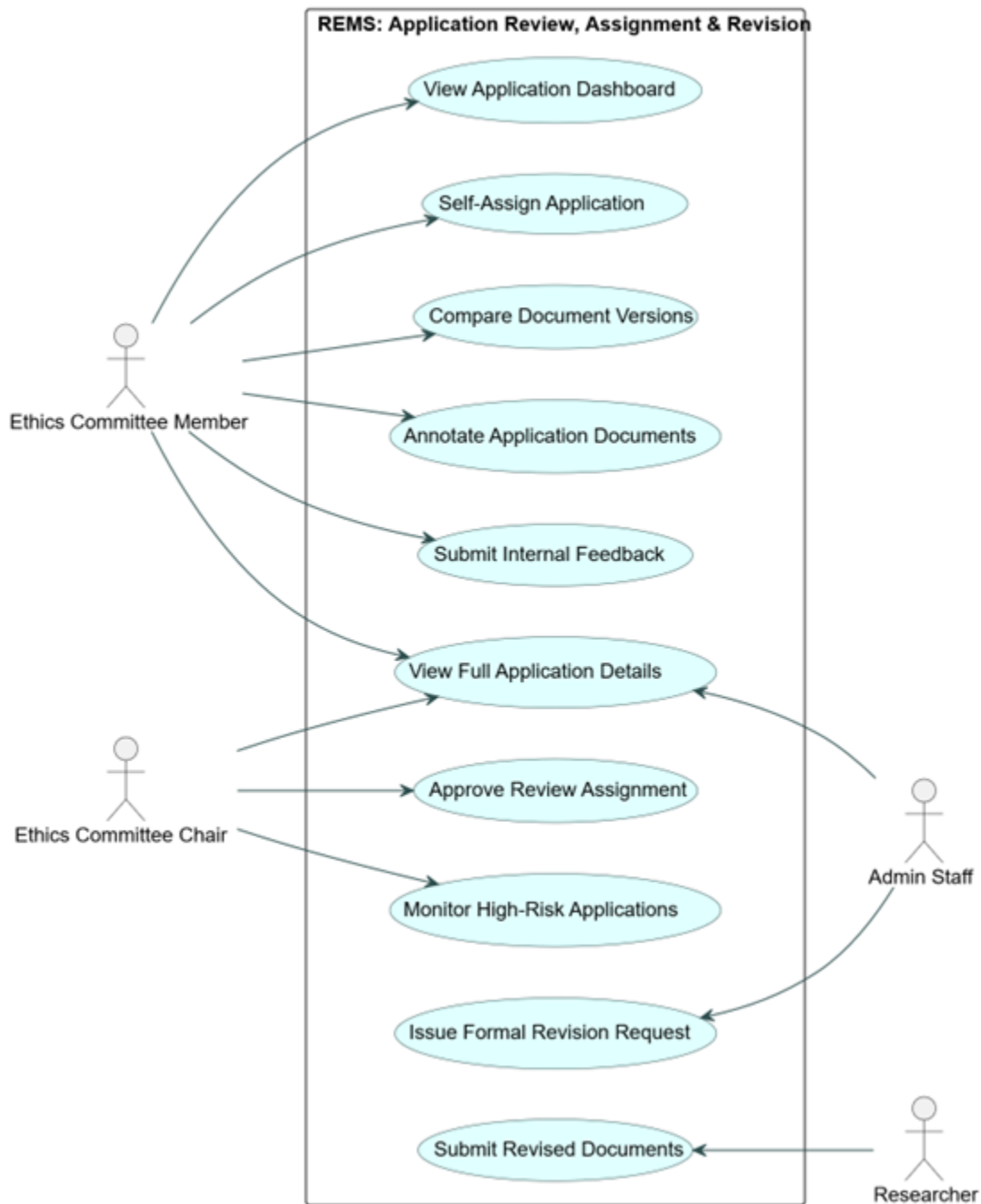
Requirements List:

Requirement ID	Type	Description
REQ-AC-01	Functional	The system shall implement Role-Based Access Control (RBAC) recognizing at least the following roles: Principal Investigator, Administrative Staff, Committee Member, and Committee Chair.
REQ-AC-02	Functional	The system shall restrict unassigned Committee Members to view only application metadata (e.g., Application Title, Status, Assigned Reviewer).
REQ-AC-03	Functional	The system shall grant full read-access to application documents, internal notes, and audit trails only to the specifically assigned Committee Member(s), the Committee Chair, and relevant Administrative Staff.
REQ-CON-01	Non-Functional	The system shall implement concurrency control mechanisms to prevent multiple users from simultaneously editing and overwriting the same application record or status.
REQ-WF-01	Functional	The system shall provide a centralized dashboard displaying a queue of all newly submitted, unassigned applications to all Committee Members.
REQ-WF-02	Functional	The system shall allow Committee Members to securely propose themselves as reviewers (self-assign) for unassigned applications from the centralized dashboard.
REQ-WF-03	Functional	The system shall require the Committee Chair to officially approve or modify proposed reviewer assignments before the application status transitions to "Under Review".

REQ-WF-04	Functional	The system shall automatically flag applications that meet predefined high-risk criteria and immediately generate a notification to the Committee Chair for direct oversight.
REQ-REV-01	Functional	The system shall allow assigned Committee Members to change an application's status to "Requires Revision" and attach internal review feedback.
REQ-REV-02	Functional	The system shall route applications flagged as "Requires Revision" by the committee to a dedicated Administrative Staff queue, preventing direct automated feedback delivery to the researcher.
REQ-REV-03	Functional	The system shall allow Administrative Staff to review internal committee feedback, format it, and issue a formalized revision request notification to the Principal Investigator.
REQ-REV-04	Functional	The system shall provide a digital annotation tool allowing assigned Committee Members to highlight and directly link their feedback to specific sections or attached documents within the application.
REQ-DOC-01	Functional	The system shall act as the single centralized repository for all application files, maintaining a unified, chronological history of all document versions.
REQ-DOC-02	Functional	The system shall automatically label the most recently uploaded document as the "Current Version" while archiving all prior submissions under a read-only historical tab.
REQ-DOC-03	Functional	The system shall allow reviewers to view previous document versions side-by-side with the current version within the same browser interface to verify changes.
REQ-NOT-01	Functional	The system shall automatically send an email reminder notification to the Principal Investigator if a requested revision or missing document remains unsubmitted after a system-configured number of days (e.g., 7 days).
REQ-US-01	Non-Functional	The system's document viewer interface shall load side-by-side version comparisons in under 3 seconds to ensure efficient user experience for reviewers.

Use Case Diagram and Use Case Description

1. Use Case Diagram



2. Detailed Use Case Description

Use Case Name: Issue Formal Revision Request

Primary Actor: Administrative Staff

Secondary Actor: Researcher

Brief Description:

This use case describes how administrative staff review internal feedback from committee members and issue a formal revision request to the researcher through the system.

Pre-conditions:

- The Administrative Staff member is authenticated and logged into the system
- An assigned Ethics Committee Member has reviewed the application and set its status to **“Requires Revision”**, with internal feedback attached
- The application appears in the Administrative Staff’s revision queue

Postconditions (Success Guarantee):

- A formal revision request has been issued to the Principal Investigator
- The application status is updated to **“Awaiting Revised Submission”**
- The action is recorded in the system audit trail, including user identity, timestamp, and request content

Main Success Scenario:

1. The Administrative Staff logs into the system
2. The system displays the dashboard with a queue of applications requiring revision
3. The Administrative Staff selects an application marked as “Requires Revision”
4. The system displays the application details along with internal reviewer comments
5. The Administrative Staff reviews the internal comments provided by the Committee Member
6. The Administrative Staff composes a structured and clear revision request based on the comments
7. The Administrative Staff submits the formal revision request through the system
8. The system sends a notification to the Researcher with the revision request
9. The system updates the application status to “Awaiting Revision”
10. The system records the action in the audit trail.

Alternative and Exceptional Flows:

Alternative A: Unclear or Incomplete Reviewer Notes

(Branches from Step 9)

- The Administrative Staff determines that the reviewer comments are unclear or insufficient
- The Administrative Staff sends a clarification request to the relevant Ethics Committee Member via the system
- The system notifies the Committee Member and places the application on hold
- The Ethics Committee Member provides updated or expanded feedback
- The Administrative Staff receives the updated feedback and resumes the process at Step 5

Alternative B: Researcher Does Not Respond Within Expected Timeframe

(Branches from Step 9)

- The system monitors the elapsed time since the revision request was issued
- If no response is received within a configured timeframe (e.g., 7 days), the system sends a reminder notification to the Researcher
- The application remains in “Awaiting Revision” status until a response is received
- The system logs the reminder in the audit trail
- If no response is received after a further defined period, the system alerts Administrative Staff for escalation or manual closure

Exceptional Flow C: System Failure During Submission

(Branches from Step 7)

1. The system fails to deliver the revision request due to a system or network error
2. The system displays an error message to the Administrative Staff and retains the drafted request
3. The Administrative Staff retries submission once the system is available
4. The system does not update the application status or audit trail until the request is successfully delivered